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Machine learning for fault detection in refrigeration systems using difluoromethane refrigerant

Tue Duy Nguyen [a], Ha Manh Bui [b,*]

[a] Faculty of Mechanical - Electrical and Computer Engineering, School of Technology, Van Lang University, Ho Chi Minh City, Viet Nam
[b] Faculty of Engineering and Technology, Saigon University, 273 An Duong Vuong Street, District 5, Ho Chi Minh City, Viet Nam

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ABSTRACT

Refrigeration systems are vital to residential and industrial cooling applications; however, faults such as refrigerant leakage and air filter clogging can severely compromise energy efficiency and reduce equipment lifespan. This study explores the potential of four machine learning (ML) models—Naïve Bayes, Generalized Linear Model (GLM), Decision Tree and Random Forest—for detecting two prevalent fault types in systems using difluoromethane (R32) as the working fluid. A synthetic dataset comprising 1998 samples was developed in Python, simulating normal operation alongside refrigerant leakage and filter clogging scenarios, based on the technical characteristics of R32 air-conditioning systems. Feature engineering and statistical visualization techniques were employed to enhance classification accuracy. All models demonstrated high predictive performance (accuracy >96 %), with Naïve Bayes achieving 100 % accuracy, indicating potential overfitting. Decision Tree and Random Forest models maintained strong generalization capabilities, with accuracies of 97.9 % and 97.4 %, respectively, suggesting practical applicability in real-time fault diagnosis. The proposed approach enables early detection of operational issues, thereby reducing energy losses and extending system service life. This research aligns with the United Nations Sustainable Development Goals (SDGs), particularly SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation and Infrastructure) and SDG 12 (Responsible Consumption and Production), by fostering intelligent and sustainable solutions for residential cooling technologies.

1. Introduction

A lack of refrigerant significantly impacts refrigeration system performance by increasing energy consumption and reducing efficiency. Low refrigerant levels diminish the system's cooling capacity, leading to longer run times, overheating and premature compressor failure. Additionally, clogged refrigerant filters contribute to increased electricity usage and reduce compressor lifespan. Both of these issues cause a drop in suction pressure, which subsequently raises the compression ratio and energy demand. Elkady et al. [1] reported that lower evaporator pressures increase power consumption, further reducing system efficiency. Notably, refrigeration systems rank among the top three electricity-consuming equipment globally, with an annual consumption of nearly 1.8 trillion kWh. Consequently, effective maintenance and fault diagnostics are crucial to mitigate energy losses and environmental impacts.

Machine Learning (ML) has been widely adopted across various industrial applications, including refrigeration systems, to enhance efficiency, predictive maintenance and energy savings. ML-based approaches diagnose faults by analyzing real-time sensor data (e.g., temperature and pressure) to detect anomalies and identify specific issues such as refrigerant leaks or compressor failures. Several studies have demonstrated the effectiveness of ML techniques in fault detection. Fan et al. [2] employed a transformer-based self-supervised learning method to improve fault diagnosis performance in HVAC (Heating, Ventilation and Air Conditioning) systems with limited labeled data. This approach effectively processes sequential data, making it suitable for time-series analysis. Hamayat et al. [3] evaluated eight ML models for HVAC fault detection, with Extra Tree, Random Forest and CatBoost outperforming others. Among them, Random Forest and CatBoost demonstrated superior efficiency in identifying individual faults. Addressing the challenge of data labeling in supervised learning, Fan et al. [4] integrated active learning with semi-supervised learning to enhance fault diagnosis while reducing annotation costs. Lu et al. [5] proposed an improved variational autoencoder (MCVAE) combined with a co-training (CT) framework to tackle insufficient training samples for chiller fault diagnosis,

* Corresponding author.
E-mail addresses: tue.nd@vlu.edu.vn (T.D. Nguyen), manhhakg@sgu.edu.vn (H.M. Bui).

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achieving superior effectiveness and efficiency compared to existing methods. Guo et al. [6] introduced four transfer learning approaches for CO_2 heat pump fault detection, demonstrating accuracy improvements from 97.79 % to 98.89 %. In contrast, the correlation alignment (CORAL) method showed a remarkable accuracy increase from 15.67 % to 74.23 %.

Alternative ML-based approaches have also been explored for HVAC fault detection. Takeuchi and Saito [7] developed a physics-informed ML method utilizing a derived scaling law for early refrigerant leak detection, validated with both laboratory and real-world data. Mei et al. [8] introduced a semi-supervised conditional Wasserstein Generative Adversarial Network (SS-CWGAN) to diagnose faults by leveraging both labeled and unlabeled data, significantly improving fault detection accuracy. Similarly, Abdollah et al. [9] implemented a transformer encoder architecture, known for its efficiency in sequential data processing, to enhance HVAC fault detection. Wang et al. [10] proposed a systematic feature selection approach for chiller fault diagnosis, integrating optimization algorithms with ML techniques to refine feature subsets and enhance diagnostic accuracy. Yabin Guo et al. [11] applied a Deep Belief Network (DBN) for variable flow refrigerant system fault diagnosis, achieving a detection accuracy of 97.7 % with an optimal network configuration.

Altair AI Studio (formerly RapidMiner) has emerged as a widely used data science platform in industrial and research applications due to its high accuracy and fast performance. This platform supports various ML techniques, including Decision Trees, Support Vector Machines and Neural Networks, facilitating tasks such as data preparation, deep learning, text mining and predictive analytics. Sujana et al. [12] demonstrated the effectiveness of RapidMiner in stock price prediction using a Neural Network model, achieving a root mean square error (RMSE) of 0.989. Similarly, Tue Duy Nguyen et al. [13] applied four ML models in RapidMiner to predict total alkalinity in saline water, reporting its suitability for such applications.

Despite its potential, Altair AI Studio has seen limited application in refrigerant fault detection. This study aims to address this gap by investigating two common faults in R32 refrigerant systems: refrigerant leakage and filter clogging. These faults are prevalent yet often misdiagnosed by service technicians. R32 refrigerant was selected due to its widespread use in air conditioning systems and its lower global warming potential (GWP). An operational dataset comprising 1998 samples was generated using Python to simulate three scenarios: normal operation, refrigerant leakage and filter clogging. Four ML models—Naïve Bayes, Generalized Linear Model, Random Forest and Decision Trees—were employed to diagnose these faults. This study contributes to the field by demonstrating the application of ML for fault detection in refrigerant systems, highlighting the potential of AI-based diagnostic tools to enhance maintenance efficiency and reduce energy consumption.

2. Methodology

This study aims to rapidly identify key fault conditions—specifically refrigerant shortage and filter clogging—in an R32 refrigerant system using classical machine learning models. The diagnostic approach is based on four measurable thermodynamic parameters: suction pressure (bar), superheat (K), subcooling (K) and condensing pressure (bar). These parameters were selected due to their sensitivity to system anomalies and their availability in standard HVAC sensor setups. The operating principle of the vapor-compression refrigeration system under study is illustrated in Fig. 1.

2.1. Refrigeration cycle and fault conditions

In a typical air-cooled refrigeration cycle, low-pressure refrigerant vapor exits the evaporator (component 5) and is drawn into the compressor (1), where it is compressed into a high-pressure, high-temperature gas. The superheat value—defined as the temperature difference between the vapor at the evaporator outlet and its saturation temperature—ensures that no liquid droplets enter the compressor, which could otherwise cause mechanical damage.

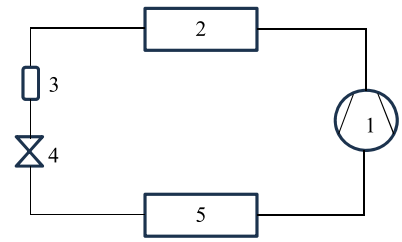

Fig. 1. Schematic diagram of the refrigeration cycle (1) Compressor, (2) Condenser, (3) Filter-drier, (4) Expansion valve and (5) Evaporator.

The superheated vapor then flows into the condenser (2), where it releases heat to the surroundings and condenses into a subcooled liquid at a relatively constant condensing pressure. The subcooling parameter, defined as the temperature difference between the condenser saturation temperature and the liquid refrigerant temperature at the condenser outlet, helps prevent flash gas formation and improves system efficiency.

After condensation, the liquid refrigerant passes through a filter-drier (3), which removes impurities and moisture that could harm system components. The refrigerant then flows through the expansion valve (4), where its pressure and temperature decrease before entering the evaporator (5) again. In the evaporator, the refrigerant absorbs heat from the environment, evaporates and the cycle repeats.

Karimi et al. [14] noted that for air-cooled condensers, the condensing temperature typically exceeds the ambient temperature by 10–15°C, while the evaporator temperature (suction pressure) is generally 10–15°C lower than the indoor air temperature. Babu et al. [15] recommended maintaining subcooling values in the range of 3–5°C to ensure the refrigerant remains in liquid form before entering the expansion device, thereby improving COP. Sunu et al. [16] found that superheat values ranging from 4.6 to 7.6°C ensured the refrigerant remained completely dry before entering the compressor.

An R32 air-cooled refrigerant system was designed to operate within a room temperature range of 16–25°C and a condensing temperature range of 40–50°C.

2.2. Data generation

A synthetic dataset comprising 1998 samples was generated to simulate three operating conditions: normal operation, refrigerant shortage and filter clogging. Each class was equally represented. The dataset was created based on the operating parameter ranges summarized in Table 1, taking reference from system design standards [14–16]. Four key features were selected as model inputs:

- Suction pressure (bar)
- Superheat value (K)
- Subcooling value (K)
- Condensing pressure (bar)

Table 1
Design and Operating Parameters of the R32 Air-Cooled Condenser Refrigerant System.

Parameters	Normal condition	Refrigerant shortage	Filter clog
Suction pressure (bar)	8.8–11	2–8.7	2–8.7
Superheat value (K)	5–10	11–25	11–25
Subcooling value (K)	3–5	0–2.9	6–15
Condensing pressure (bar)	25–32	18–24.5	32.5–40

To develop the classification dataset, synthetic data were generated using a uniform sampling strategy within predefined parameter ranges informed by domain expertise and established system behavior patterns. For each of the three operating conditions—normal operation, refrigerant shortage and filter clogging—666 data points were created to ensure class balance, resulting in a total of 1998 samples. Four key thermodynamic parameters were considered: suction pressure, superheat, subcooling and condensing pressure. These variables were selected due to their sensitivity to refrigerant faults and their widespread availability in HVAC sensor systems.

The parameter ranges for each fault condition were defined as follows:

- Normal operation: suction pressure (8.8–11 bar), superheat (5–10 K), subcooling (3–5 K) and condensing pressure (25–32 bar), reflecting typical system performance under stable load.
- Refrigerant shortage: suction pressure (2–8.7 bar), superheat (11–25 K), subcooling (0–2.9 K), with condensing pressure in the normal range. These symptoms reflect reduced refrigerant mass flow, resulting in lower suction pressure and increased superheat.
- Filter clogging: suction pressure (2–8.7 bar) and superheat (11–25 K), similar to refrigerant shortage, but combined with elevated condensing pressure (32.5–40 bar) and increased subcooling (6–15 K) due to refrigerant accumulation upstream of the restriction.

By ensuring non-overlapping thresholds in condensing pressure and subcooling between the two fault classes, the model can distinguish between similar fault symptoms. No artificial noise was injected into the dataset to maintain controlled conditions and isolate the influence of each parameter on classification outcomes. After generation, all samples were randomly shuffled to eliminate ordering bias. A pseudocode summarizing the data generation procedure is provided in Appendix A to facilitate transparency and reproducibility. This approach supports rapid prototyping of diagnostic algorithms when field data are unavailable or difficult to label.

2.3. Data preparation

Table 1 displays the values of the refrigerant system under normal conditions and faulty conditions. Evaporator and condenser pressures are measured in bar units and correspond to their respective saturation temperatures. In real-life applications, the values in the table depend on operating conditions, but the condensing pressure should not be excessively high to avoid potential hazards.

Moreover, it can be observed that both faulty conditions are characterized by lower suction pressure and higher superheat values. These symptoms can confuse technicians in identifying whether the issue is due to refrigerant shortage or a clogged filter. The key distinguishing factors are the subcooling value and condensing pressure. When a refrigerant shortage occurs, the subcooling value and condensing pressure decrease due to insufficient refrigerant in the system, whereas a clogged filter causes the subcooling value and condensing pressure to increase because liquid refrigerant accumulates before the filter, causing its temperature to gradually decrease, meaning that subcooling value increase and condensing pressure to increase rapidly.

Based on the assumed values in Table 1, a total of 1998 data points were generated, with three corresponding symptom categories—normal condition, lack of refrigerant and filter clog—equally divided using Python code.

2.4. Machine learning models

Using 1998 data samples (1428 for traning and 570 for testing), machine learning classification models such as Naïve Bayes, Generalized Linear Model, Random Forest, Decision Trees were applied to compare classification results. The data analysis was conducted using Rapid Miner platform and the analysis procedure is outlined in Fig. 2.

To ensure that the data is in an optimal format for machine learning models, data preparation was performed. In this step, missing or inconsistent values were addressed and the dataset was split into training and testing subsets. Specifically, 1428 samples of the data were used for training, while the remaining 570 samples were allocated for testing.

Four machine learning classification models were applied, including Naïve Bayes, Generalized Linear Model (GLM), Random Forest and Decision Trees. These models are widely used for classification in various tasks. Naïve Bayes, known for its probabilistic classification, is commonly used in fault detection systems and medical diagnosis. Pratiwi & Hamami [17] stated that the Naïve Bayes model was chosen in their study on the classification of junior high school mathematics questions due to its simplicity and ability to process data efficiently. Argirusis et al., [18] applied Random Forest in an IoT-based refrigerant gas classification system for environmental monitoring, highlighting its high accuracy. Furthermore, this model enhances diagnostic accuracy in disease classification, as demonstrated in the study by Badhoutiya et al., [19]. Generalized Linear Models (GLMs) are an algorithmic extension of standard regression techniques to handle non-normal distributions. Therefore, they can be used for predicting environmental impacts and medical outcomes, such as asthma hospitalizations [20]. For obesity risk prediction, Putri et al., [21] utilized an obesity classification dataset containing 2111 records and 17 features obtained from Kaggle. Their study reported that Decision Trees achieved the highest accuracy at 84.98 %, followed by K-Nearest Neighbors (KNN) at 83.55 %. Naïve Bayes ranked third with an accuracy of 77.48 %, while Support Vector Machine (SVM) had the lowest accuracy at 77.32 %.

2.5. Model evaluation

Model performance was assessed using the following metrics:

- Accuracy: Measures the overall correctness of the model; suitable for balanced datasets.
- Precision: Emphasizes minimizing false positives; important in applications requiring high reliability.
- Recall: Focuses on minimizing false negatives; critical in early fault detection.
- Confusion Matrix: Provides a detailed breakdown of true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN) for a comprehensive evaluation.

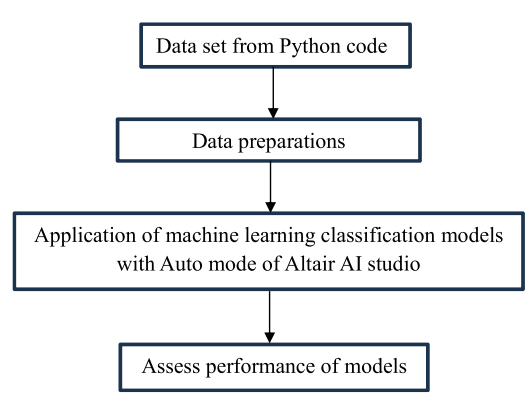

Fig. 2. Procedure for Machine learning.

3. Results and discussion

3.1. Data visualization

Histograms, as shown in Fig. 3, were used to analyze the distribution of data, illustrating how data points are spread across different ranges and helping to identify issues such as outliers, skewness and data gaps.

The histograms in Fig. 3 show the distribution of four key variables in the refrigeration system: suction pressure, superheat, subcooling and condensing pressure. Each variable exhibits a distinct distribution pattern.

Suction pressure follows a fairly uniform distribution, suggesting stable variations without significant skewness. Superheat exhibits a bimodal distribution, indicating two separate operating conditions—one corresponding to normal functioning and the other linked to potential failures, such as refrigerant loss or filter clogging.

The subcooling histogram is right-skewed, with most values concentrated in the lower range (0–4 K) and the highest frequency occurring at 4 K. This occurs because normal operation and refrigerant shortage conditions typically fall within 0–5 K, whereas higher subcooling values (6–15 K), which are associated with filter clogging, occur less frequently. As a result, this imbalance creates a skewed tail in the histogram. Furthermore, Table 1 confirms that subcooling values above 6 K predominantly occur in filter clogging conditions, though these cases are less frequent than those in normal and refrigerant shortage scenarios.

Condensing pressure is evenly distributed, but certain peaks suggest potential system inefficiencies at specific pressure levels. In general, the presence of skewness in superheat and subcooling suggests that external factors, such as refrigerant levels or system blockages, influence the refrigeration system's performance. Therefore, it is essential to implement anomaly detection and predictive maintenance strategies to identify and prevent potential failures. The information regarding the data is listed in Table 2.

Table 2 provides an overview of key statistical properties of the dataset including missing values, average values, maximum and minimum values corresponding to each parameter. Additionally, the standard deviation is included to assess the variability of the data. In term of type values, suction pressure, superheat, subcooling and condensing pressure are real-valued, while label is nominal (categorical). The table shows that there are no missing values (0) across all parameters, meaning the dataset is complete and does not require imputation. Among the input features, condensing pressure exhibits the highest standard deviation (6.482), followed by superheat (6.116). Subcooling ranks third, with a deviation of 4.155, while suction pressure (evaporator pressure) has the lowest deviation, at 2.709. A higher deviation indicates greater variability in the data, whereas a lower deviation suggests more consistency in the values. The variability in data arises due to the distribution of parameters across the three conditions (Normal, Refrigerant Shortage and Filter Clog). For instance:

a/ Condensing pressure drops to 18–24.5 bar under refrigerant shortage, while it rises to 32.5–40 bar in filter clogging conditions.

b/ Superheat values fluctuate significantly, ranging from 11 to 25 K under refrigerant shortage compared to 5–10 K in normal conditions.

Moreover, the classification labels representing the three conditions (Normal, Lack of Refrigerant and Filter Clog) are evenly distributed (666 samples per label), ensuring optimal balance. Furthermore, no extreme outliers are present in the data. Therefore, the data quality is good for data analysis and machine learning applications.

3.2. Correlation between parameters

Correlation describes the relationship between two variables, indicating how changes in one may relate to changes in the other. While correlation helps identify features that may contribute to a target variable, it only demonstrates an association, not a causal relationship. The correlations between variables are shown in Fig. 4.

Fig. 4 presents the correlation matrix for key variables in the

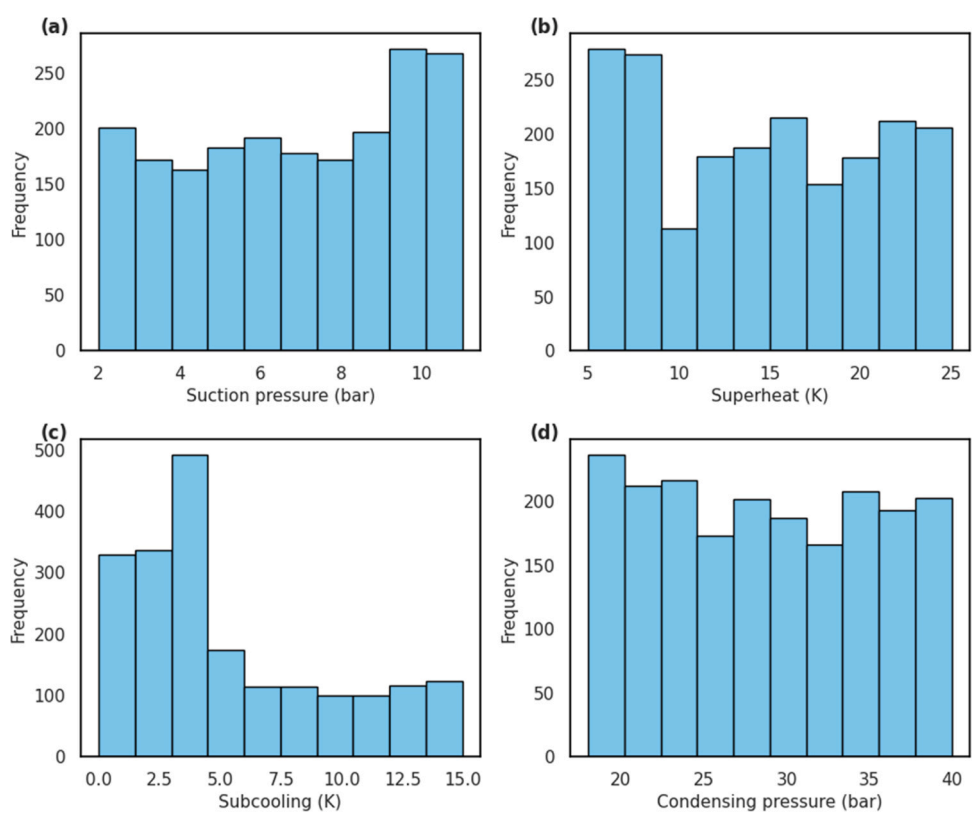

Fig. 3. Histogram of parameters: a) Suction pressure, b) Superheat, c) Subcooling, d) Condensing pressure.

Table 2
Information regarding the data.

Parameter	Type	Missing	Min	Max	Average	Deviation
Suction_pressure	Real	0	2	10.999	6.832	2.709
Superheat	Real	0	5.004	24.992	14.563	6.116
Subcooling	Real	0	0	14.959	5.342	4.155
Condensing_pressure	Real	0	18.002	39.959	28.671	6.482
Label	Nominal	0	normal (666)	filter clog (666)	filter clog (666), lack of refrigerant (666), normal (666)	-

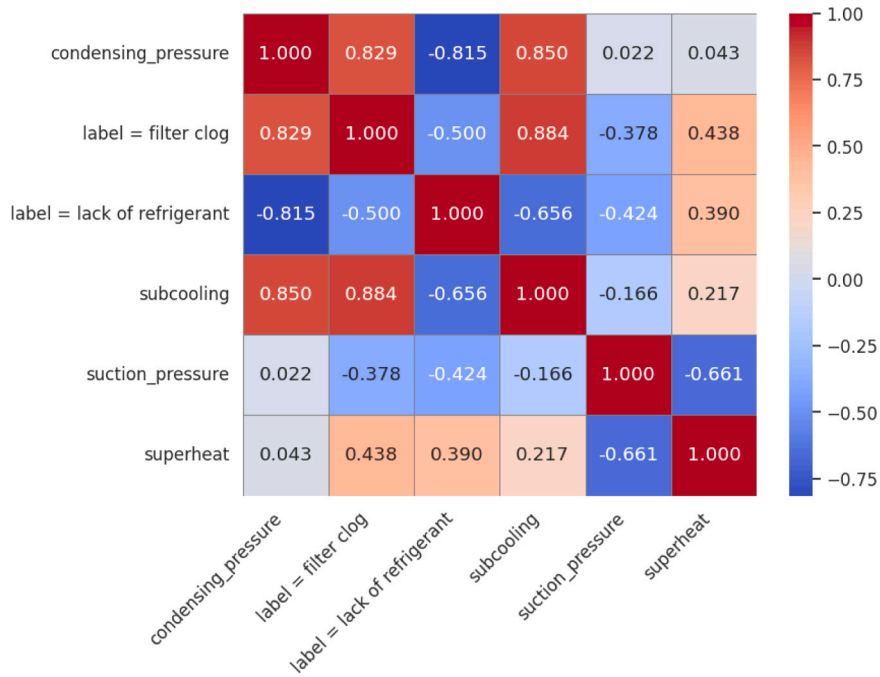

Fig. 4. The correlations between variables.

refrigeration system dataset, including condensing pressure, superheat, suction pressure, subcooling and the categorical labels representing fault types (filter clog and lack of refrigerant). This matrix helps reveal the strength and direction of linear relationships between system parameters and fault conditions.

Focusing on the filter clog scenario, strong positive correlations are observed with condensing pressure (r = 0.829) and subcooling (r = 0.884), while superheat shows a moderate positive correlation (r = 0.438) and suction pressure displays a moderate negative correlation (r = –0.378). These results indicate that, when a filter clog occurs, condensing pressure, subcooling and superheat tend to increase, whereas suction pressure tends to decrease.

In terms of refrigerant shortage, it has a strong negative correlation with condensing pressure (-0.815) and subcooling (-0.656), as well as a moderate negative correlation with suction pressure (-0.424). This indicates that this type of fault causes a decrease in condensing pressure, suction pressure and subcooling. Conversely, it has a moderate positive correlation with superheat (0.390), suggesting that this fault leads to an increase in superheat.

3.3. Model accuracy visualization

Accuracy is a metric that evaluates the overall correctness of a classification model. It is calculated by dividing the number of correct predictions by the total number of predictions. This shows how well the model can classify examples correctly out of all the data. A higher accuracy means the model is better at identify the difference between different classes.

The parameter accuracy of four models such as Naïve Bayes, Generalized Linear Model, Random Forest, Decision Trees are listed in Table 3.

Table 3 compares the performance of four models with key parameters including accuracy, standard deviation, gains, training time and scoring time. Regarding accuracy and standard deviation, Naïve Bayes achieves the highest accuracy (100 %) with lowest deviation (0 %), but it is careful when applying in real life because a perfect accuracy often suggests potential overfitting, which needs further verification. Followed by Decision Trees (97.9 %) and Random Forest (97.4 %) which are also perform very well, with a low standard deviation (~1.1 %-1.2 %), indicating stable performance across different training/testing splits. While Generalized Linear Model has the lowest accuracy (96.1 %) with highest deviation (± 7.2 %) making it less reliable than the other models.

Regarding the gain parameter, the Naïve Bayes model achieves the highest value at 746, followed by Decision Trees (722) and Random Forest (716). This suggests that the Naïve Bayes model has the potential to be the most useful in terms of ranking or scoring outcomes. The Generalized Linear Model ranks last, with a gain parameter of 700, which corresponds to the lowest accuracy among the four models. Additionally, it is observed that the gain parameter correlates with both accuracy and deviation.

In terms of training time, Random Forest (41 ms) and Decision Trees (46 ms) are the fastest to train, as these models are optimized for speed. They are followed by Naïve Bayes (109 ms), while the Generalized Linear Model (GLM) takes the longest training time at 341 ms.

Regarding scoring time (inference speed), Decision Trees are the fastest (150 ms), making them suitable for real-time applications. They are followed by GLM (199 ms), which is a good choice for fast

Table 3
Accuracy of machine learning models for classification.

Model	Accuracy	Standard Deviation	Gains	Training Time (milliseconds)	Scoring time (milliseconds)
Naive Bayes	100.00 %	± 0.0 %	746	109	679
Generalized Linear Model	96.10 %	± 7.2 %	700	341	199
Random Forest	97.4 %	± 1.1 %	716	41	327
Decision Trees	97.9 %	± 1.2 %	722	46	150

predictions that do not require highly complex computations. Random Forest (327 ms) is slightly slower, as it aggregates results from multiple decision trees. Naïve Bayes has the slowest scoring time (679 ms), requiring more computational resources during prediction, likely due to the complexity of handling multiple probability distributions.

Overall, Naïve Bayes demonstrates perfect accuracy (100 %) but has the longest scoring time (679 ms), which may indicate overfitting or computational inefficiency. If Naïve Bayes' perfect accuracy is not due to overfitting, it could be valuable in applications where high precision is critical, but further analysis is necessary. Decision Trees offer a strong balance between high accuracy (97.9 %), low standard deviation (1.2 %), fast training time (46 ms) and the lowest scoring time (150 ms), making them highly efficient. Random Forest is also highly accurate (97.4 %) but has a longer scoring time (327 ms), which may not be ideal for real-time applications. The Generalized Linear Model (GLM) has the lowest accuracy (96.1 %) and the highest variability (±7.2 %), making it less reliable than the other models. However, it has a relatively low scoring time (199 ms). Therefore, unless simplicity is preferred over accuracy, it should be reconsidered.

In refrigerant fault identification, diagnostic accuracy is the top priority, whereas quick response time is not the main concern. Therefore, models such as Naïve Bayes, Decision Trees and Random Forest will continue to be analyzed in the next steps to gain deeper insights.

3.4. Confusion matrix, class recall and class precision

Altair AI Studio splits the data into training and testing sets, with 1428 samples in the training set and 570 samples in the testing set. In addition to accuracy, other evaluation metrics, including the confusion matrix, class recall and class precision, also need to be assessed. The confusion matrices for three models—Naïve Bayes, Decision Trees and Random Forest—are presented in Table 4.

Table 4 depict key parameters, including the confusion matrix, class precision and class recall of three models. Based on the confusion matrix table, class precision and class recall, the Naïve Bayes model achieves the highest accuracy of 100 % without any misclassification, making it an ideal model. However, because of its perfect accuracy, it should be reconsidered to avoid potential overfitting.

The Decision Tree model follows with a slight error, misclassifying six instances of "Filter Clog" and six instances of "Lack of Refrigerant" as "Normal." Despite this, it maintains high precision (94.15 % for "Normal," 100 % for other classes) and high recall (96.67 % for "Filter Clog," 96.95 % for "Lack of Refrigerant").

Table 4
Confusion matrix, class precision and class recall for the models.

Model	Predicted / Actual	True Normal	True Filter Clog	True Lack of Refrigerant	Class Precision
Naïve Bayes	Normal	188	0	0	100 %
	Filter Clog	0	185	0	100 %
	Lack of Refrigerant	0	0	197	100 %
	Class Recall	100 %	100 %	100 %	-
Decision Tree	Normal	193	6	6	94.15 %
	Filter Clog	0	174	0	100 %
	Lack of Refrigerant	0	0	191	100 %
	Class Recall	100 %	96.67 %	96.95 %	-
Random Forest	Normal	188	8	7	92.61 %
	Filter Clog	0	177	0	100 %
	Lack of Refrigerant	0	0	190	100 %
	Class Recall	100 %	95.68 %	96.45 %	-

The Random Forest model exhibits a slightly higher misclassification rate, with eight instances of "Filter Clog" and seven instances of "Lack of Refrigerant" incorrectly classified as "Normal." As a result, the model achieves perfect precision (100 %) for both fault classes but slightly lower recall rates—95.68 % for "Filter Clog" and 96.45 % for "Lack of Refrigerant." For the "Normal" class, precision drops to 92.61 %. In comparison, the Decision Tree model offers more balanced recall across all classes, suggesting a better differentiation capability, particularly between normal and fault conditions.

Interestingly, the Naïve Bayes model attains perfect classification accuracy on the current dataset. While this outcome appears promising, it also raises concerns regarding potential overfitting, especially given the synthetic nature of the data. This suggests the model may have memorized statistical patterns specific to the current data partition, thus limiting its generalizability. Consequently, additional validation on unseen or real-world data is critical to assess model robustness.

Despite strong classification performance by both tree-based models, their learning mechanisms differ fundamentally. Decision Trees, while easy to interpret and fast to train, are known to be prone to overfitting, particularly when trees are allowed to grow without pruning [22]. In contrast, Random Forest leverages an ensemble of decision trees, each trained on bootstrapped subsets of the data and random feature selections. This aggregation mechanism reduces model variance and enhances generalization, particularly in the presence of noise or incomplete patterns. These attributes make Random Forest a more reliable choice for deployment in real-world fault diagnosis scenarios [23].

Supporting this perspective, Hartama and Amalya [24] found in their comparative analysis of Decision Tree, ID3 and Random Forest that the ensemble method consistently provided more stable predictions and was less susceptible to overfitting. In light of these findings, Random Forest may offer a favorable balance between interpretability and robustness, especially in operational environments with sensor noise and dynamic conditions.

To further validate model performance and mitigate overfitting risks, future research should incorporate k-fold cross-validation to evaluate consistency across multiple data splits. Additionally, applying the models to real operational datasets would enhance practical credibility. A broader suite of evaluation metrics—such as Receiver Operating Characteristic (ROC) curves, Area Under the Curve (AUC) and learning curves—should also be employed. Platforms such as Python's scikit-learn or R provide the flexibility to conduct such analyses systematically, allowing a more comprehensive assessment of model behavior.

3.5. Feature importance analysis

Given that both Decision Tree and Random Forest are inherently interpretable models, feature importance analysis was performed to understand which sensor variables most influenced classification decisions. This analysis not only explains model behavior but also provides practical insights for system monitoring and maintenance.

In the Random Forest model, suction pressure emerged as the most critical feature (importance score: 0.183), followed by superheat

(0.089). Condensing pressure (0.017) and subcooling (0.015) were found to have relatively lower impacts on fault detection. Similarly, in the Decision Tree model, suction pressure remained the dominant feature (0.176), with superheat (0.044), subcooling (0.041) and condensing pressure (0.022) contributing to a lesser extent.

These results highlight the diagnostic value of suction pressure and superheat in distinguishing abnormal conditions in R32 systems. As both fault types—refrigerant shortage and filter clogging—can exhibit similar signatures in these parameters, secondary features like condensing pressure and subcooling become essential for differentiating between fault types. This reinforces the importance of multivariate analysis in HVAC diagnostics.

By identifying key sensor inputs, such analysis supports informed decision-making in field diagnostics, allowing technicians to prioritize the most relevant parameters during real-time fault detection. Table 5 summarizes the feature importance values for both models, offering a visual reference for interpreting the model logic.

3.6. Performance comparison with previous studies

To contextualize the effectiveness of the proposed machine learning models for fault detection in R32 refrigerant systems, Table 6 presents a comparative summary of recent studies employing various algorithms for similar diagnostic purposes. Rather than isolated reporting, the following discussion highlights how the results from the current study align with or differ from existing works, thereby underlining both the practical relevance and the novelty of this research.

The current study demonstrates comparable and in some cases superior, performance relative to more complex models. While deep learning approaches such as those by Wang et al. [26] and Sulaiman et al. [25] achieved high accuracy (up to 99.66 %), they often require large datasets and high computational power. In contrast, the Naïve Bayes model in this study reached 100 % accuracy on the synthetic dataset, with all other models exceeding 96 %, underscoring the efficiency of simpler classifiers when applied to well-structured fault categories like filter clogging and refrigerant shortages.

The multi-layer perceptron (MLP) model used by Boahen et al. [27] also demonstrated strong performance in detecting refrigerant charge faults, with lower mean squared errors than traditional regression methods. This suggests that even minor fault levels can be effectively modeled using nonlinear learning techniques—an insight supported by the current results, where classical models achieved high accuracy in binary fault classification.

Moreover, Soltani et al. [28] and Ebrahimifakhar et al. [29] reported accuracies in the 83–95 % range using models such as SVM, KNN and logistic regression. Although suitable for multi-fault classification, these models showed relatively lower accuracy compared to the Naïve Bayes and Decision Tree classifiers applied in this study. Notably, Zhang et al. [30] and Guo et al. [6] addressed system heterogeneity and environmental variability through ensemble and domain adaptation methods, achieving accuracies above 90 %. However, these approaches often involve more complex workflows and extensive parameter tuning.

Despite these promising findings, it is acknowledged that the current evaluation is based on synthetic data. Real-world HVAC systems exhibit noise, nonlinearity and temporal dynamics that can affect model generalizability. As Zhang et al. [30] and Guo et al. [6] demonstrated, validation using operational data or open-access datasets is essential to ensure practical deployment. Future work will therefore focus on implementing these models in embedded systems and validating them with real-time sensor inputs.

Table 5
Feature importance weights for Random forest and Decision tree models.

Attribute	Random forest	Decision tree
Subcooling	0.015	0.022
Condensing pressure	0.017	0.041
Superheat	0.089	0.044
Suction pressure	0.183	0.176

Table 6
Comparative analysis of machine learning models for fault detection of refrigerant system.

Reference	Methodology	Research Focus	Results
Sulaiman et al., [25]	Deep learning, SVM and MLP	Fault detection in air conditioning systems	MLP achieved highest accuracy and precision; overall > 94 % accuracy
Wang et al., [26]	Deep learning-based model	Fault detection and diagnosis in refrigeration systems	Accuracy ≈ 99.66 %
Boahen et al., [27]	MLR and MLP	Refrigerant charge fault detection	MLP: ±3.7 % MSE; MLR: ±4.5 % MSE
Soltani Z. et al., [28]	SVM	Classification of 20 industrial refrigeration faults	Accuracy: 95 %
Ebrahimifakhar et al., [29]	KNN, Logistic Regression, Random Forest	Fault detection in rooftop air conditioning units	Accuracy: Logistic Regression (93.6 %), RF (88.3 %), KNN (83.6 %)
Zhang et al., [30]	SVM, RF, KNN, ANN, GBDT	Fault detection for heat pump systems	Accuracy: ANN (90 %), RF (87.5 %), SVM (85 %), KNN (82 %)
Guo et al. [6]	TCA, JDA, CORAL, BDA	Fault detection of CO_2 heat pump	Accuracy: 97.79–98.89 % (TCA, JDA, BDA); CORAL: 15.67–74.23 %
Current study	Naïve Bayes, GLM, Decision Tree, Random Forest	Fault detection for R32 systems (filter clogging, refrigerant shortage)	Accuracy: NB (100 %), GLM (96.1 %), DT (97.9 %), RF (97.4 %); Precision and recall > 96 %

Importantly, the choice of classical algorithms in this study is not incidental. Models like Naïve Bayes, GLM, Decision Tree and Random Forest are computationally efficient, interpretable and fast to deploy—making them highly practical for embedded diagnostics in smart HVAC systems. While algorithms like XGBoost offer state-of-the-art performance in many domains, their complexity, sensitivity to hyperparameters and higher resource requirements often render them less favorable in time-critical or resource-constrained environments [31].

This study not only confirms the feasibility of classical models for fault detection in R32-based refrigerant systems but also contributes practical insights toward building efficient, low-cost diagnostic solutions. These findings serve as a bridge between academic research and real-world implementation, supporting future work on predictive maintenance and energy optimization in HVAC technologies.

4. Conclusion

This study demonstrates the effectiveness of classical machine learning algorithms in detecting two critical fault conditions—refrigerant shortage and filter clogging—in an R32-based air-cooled refrigeration system. Among the models evaluated, Random Forest exhibited the highest classification accuracy and superior generalization capability, attributed to its ensemble structure. Despite being trained on a limited and synthetically generated dataset, the models achieved high diagnostic reliability when informed by domain-specific knowledge, underscoring the potential of structured data combined with expert-driven feature engineering.

The selection of interpretable and computationally efficient

algorithms, such as Naïve Bayes, Generalized Linear Model and Decision Tree, further supports their applicability in real-world engineering contexts, particularly for embedded implementations and rapid deployment. Nonetheless, the study is subject to several limitations, including the reliance on synthetic fault data and a narrow set of fault categories. Moreover, model robustness under noisy or time-varying signal conditions was not assessed, as the current platform (RapidMiner) lacks built-in support for noise injection and signal degradation simulation. Addressing these gaps will be a focus of future work. Ongoing efforts will include the validation of these models using real-time IoT sensor data collected from operational air-conditioning units, expansion to a broader fault taxonomy and the integration of adaptive learning algorithms capable of capturing dynamic system behavior. These improvements are essential for advancing toward robust, real-time fault detection and predictive maintenance systems. The proposed approach aligns with global sustainability targets, particularly the United Nations Sustainable Development Goals, by contributing to energy efficiency (SDG 7), promoting innovation in industrial systems (SDG 9) and encouraging sustainable operation and maintenance practices (SDG 12). Future research will also leverage more flexible environments such as Python to simulate data uncertainty and assess model resilience under real-world conditions, thereby enhancing the scalability and robustness of the solution.

CRediT authorship contribution statement

Ha Manh Bui: Writing – review & editing. **Tue Duy Nguyen:** Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Pseudocode for Synthetic Data Generation

Initialize an empty dataset
For each fault category in ["normal", "lack of refrigerant", "filter clog"]:
Define specific parameter ranges for:
- Suction pressure
- Superheat
- Subcooling
- Condensing pressure
For i = 1–666:
Generate random values for each parameter using uniform sampling within the defined range
Assign the corresponding fault label to the data point
Append the data point to the dataset
After generating all data points:
Combine data from all categories
Shuffle the dataset to eliminate any ordering bias
Export the final dataset to a CSV file

Data availability

The data that has been used is confidential.

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